

Generate, Retrieve, or Calculate?

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Missing and Unbalanced Examples

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Similarity

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Separate a correct sum from a correct answer for Generation retrieval and calculation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Generation retrieval and calculation. Proposed explanation: “Retrieving a record proves it is current”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Trace a new library workshop

Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3. A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” Retrieve the current day and room, calculate the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose generation, retrieval or calculation for each. Which outputs need checking?

6. Solve the given case

A learner needs a new poem, $18+7$, and a passage from the class handbook.

Choose the operation for each.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Choose a method for each task: invent a dragon story; find the library's posted hours; compute 7 plus 5. Explain why the methods differ.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 25 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Label every sorter component as generate, retrieve, calculate, fixed rule, learned prediction, or human decision.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose generation, retrieval or calculation for each. Which outputs need checking?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Name the operation and its evidence

For each task name the main operation and a suitable check: find the room listed in N2; work out six tables of four places minus three reserved places; suggest an inviting title for the event. Can an application combine these operations?

My reasoning and evidence

13. W2 Separate a correct sum from a correct answer

A calculator correctly returns 24 for 6×4 . A pupil reports this as the number of non-helper places available. Explain the problem and write the correct expression, result and meaning.

My reasoning and evidence

14. W3 Inspect a record and a draft

N1 lists Room 2. N2 explicitly replaces N1 for this event and lists Wednesday in Room 4. A draft says Thursday in Room 2 with free snacks. Identify the supported corrections and the unsupported detail. Explain why finding N1 successfully would not settle the current-room question.

My reasoning and evidence

15. W4 Trace a new library workshop

Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3. A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” Retrieve the current day and room, calculate the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

My reasoning and evidence

16. W5 Do not infer a method from its output

Two applications both display “26 places remain.” Only one provides record B and a trace showing $5 \times 6 - 4$. What can and cannot be concluded from their matching sentences? Describe how you would check the numerical claim without relying on either application’s assurance.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Generate, Retrieve, or Calculate?

A fictional craft invitation combines three different operations.

Approved record C: Tuesday, Hall 4, three tables with four learner places each.

Two places are reserved for helpers; no snack information is supplied.

Generate, Retrieve, or Calculate?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieve facts

Read Tuesday, Hall 4 and 3 x 4 places from approved C.



2. Calculate

12 total places minus 2 reserved = 10 learner places.



3. Generate wording

Draft a friendly invitation using only those supported details.



4. Check claims

A sentence promising snacks has no supporting record.

CHANGE ONE THING

Reserve four places instead of two, keeping the room and day unchanged.

Explain the changed result using the matching lesson.

Label each arrow with its operation. Can the system add snacks because invitations often mention them?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Use the sorter’s approved catalogue for facts, explicit calculations for scores and checked prose for explanations.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

Use the sorter’s approved catalogue for facts, explicit calculations for scores and checked prose for explanations.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft invitation combines three different operations.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A sentence promising snacks has no supporting record.

1. Retrieve facts
2. Calculate
3. Generate wording
4. Check claims

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check claims A sentence promising snacks has no supporting record. Retrieval copies relevant source information; calculation follows arithmetic; generation proposes wording. These jobs can cooperate, but one does not validate the others. A friendly generated sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

Practice 2: explain the deciding evidence

Changed-case practice: Reserve four places instead of two, keeping the room and day unchanged. Which conclusion is supported by the supplied facts?

1. Eight learner places remain.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Eight learner places remain. $3 \times 4 - 4 = 8$; this changes capacity for learners, not the retrieved date or location.

Repair a missing building block

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields. Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap? Point to the target groups.. Return to this week and explain what you changed.

Review Similarity (ai-1/19). Goal: Calculate similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
